ELG3136 Electronics II: Lab 3 Pre-lab

Frequency Response of a Common-Emitter BJT Amplifier

Last name	First name	Student ID

Values used. $V_{CC} = 15$ V, $R_1 = 22$ kΩ, $R_2 = 12$ kΩ, $R_C = 3.3$ kΩ, $R_E = 4.3$ kΩ, $R_L = 4.7$ MΩ, $C_1 = C_2 = C_E = 10$ µF (C_1 input coupling, C_2 output coupling, C_E emitter bypass; C2, C3 and C4 in the manual's Figure 2). 2N3904: $\beta = 150$ (assumed; use your measured β), $V_{BE} = 0.7$ V, $V_T = 26$ mV, $V_A \approx 74$ V, $r_b \approx 10\ \Omega$.

Operating point

$$V_{Th} = 15 \cdot \frac{12}{22+12} = 5.29 \text{ V}, \quad R_B = R_1 \parallel R_2 = 7.76 \text{ k}\Omega$$

$$I_B = \frac{V_{Th} - V_{BE}}{R_B + (\beta+1)R_E} = 6.99 \text{ µA}, \quad I_C = \beta I_B = 1.05 \text{ mA}$$

$$V_C = 11.54 \text{ V}, \quad V_E = 4.54 \text{ V}, \quad V_{CE} = 7.0 \text{ V} \quad \text{(forward active)}$$

$$g_m = \frac{I_C}{V_T} = 40.3 \text{ mS}, \quad r_\pi = \frac{\beta}{g_m} = 3.72 \text{ k}\Omega, \quad r_o = \frac{V_A}{I_C} = 70.6 \text{ k}\Omega, \quad R_L' = R_C \parallel R_L \parallel r_o = 3.15 \text{ k}\Omega$$

I_C barely depends on β (1.04 mA for $\beta = 100$), so the estimates below are robust to the β assumption.

Task 1: Circuit with BJT parasitic capacitances

Figure 1 adds the transistor's internal capacitances (red, dashed): C_π between base and emitter (diffusion plus B-E junction capacitance) and C_μ between base and collector (B-C junction capacitance). The collector-substrate capacitance C_{CS} is negligible for a discrete 2N3904.

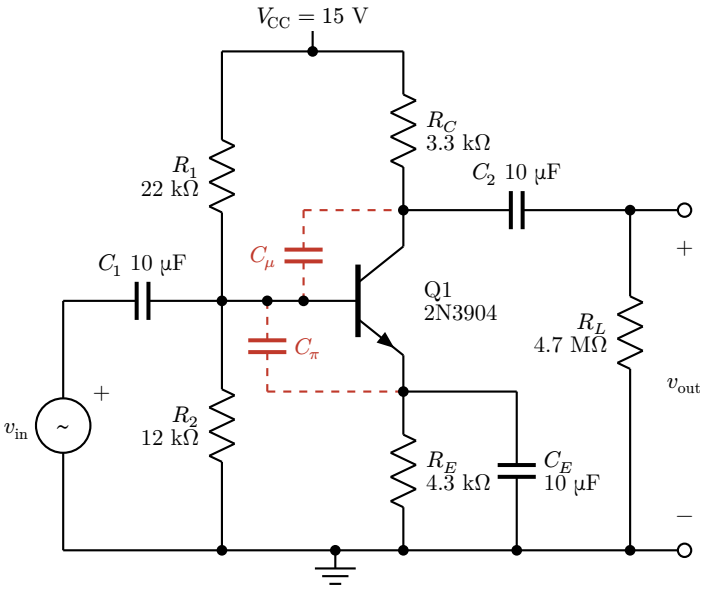


Figure 1: Common-emitter amplifier with the BJT parasitic capacitances C_π and C_μ added.

Task 2: Gain as $f \to 0$

At DC, C_1, C_2 and C_E are open circuits: C_1 blocks the signal from the base and C_2 blocks it from R_L, so $|A_v| \to 0$ ($-\infty$ dB). Each coupling capacitor contributes a zero at $s = 0$, so below every pole the slope is +40 dB/dec. Going down from mid-band, C_E opens first; with C_1 and C_2 still acting as shorts the emitter is unbypassed and

$$A_v = -\frac{\beta R'_L}{r_\pi + (\beta+1)R_E} = -0.72 \approx -\frac{R'_L}{R_E} \quad (-2.8 \text{ dB}),$$

after which C_1 (near 2 Hz) and C_2 (near 3 mHz) take the gain to zero.

Task 3: Gain as $f \to \infty$

At very high frequency C_π and C_μ become short circuits. C_π shorts v_π, so $g_m v_\pi \to 0$, and C_μ ties the collector to the internal base, which C_π and C_E hold at AC ground. Hence $v_{\text{out}} \to 0$ and $|A_v| \to 0$. With two high-frequency poles and one right-half-plane zero, the final roll-off is -20 dB/dec.

Task 4: Mid-band gain

Short C_1, C_2, C_E and open C_π, C_μ:

$$A_m = \frac{v_{\text{out}}}{v_{\text{in}}} = -g_m(R_C \parallel R_L \parallel r_o) = -(40.3 \text{ mS})(3.15 \text{ k}\Omega) \approx -127 \quad (42.1 \text{ dB})$$

Neglecting r_o gives -133. With an ideal source (MultiSim), or with v_{in} measured at the base (lab), the generator resistance does not enter. Section II of the manual quotes about -50; for this fully bypassed stage the small-signal model predicts about -127, so confirm the value in MultiSim.

Table 1: Mid-band voltage gain (manual Table 2).

Theoretical gain	**Measured gain A_m**
-127 (42.1 dB)	

Tasks 5 to 7: MultiSim model and pole-zero analysis

Insert MultiSim schematic screenshot here

Figure 2: Circuit of the manual's Figure 2 built in MultiSim (Task 5).

The hand estimates below assume an ideal source ($R_s = 0$), as in MultiSim, and are listed in Table 2 for checking the MultiSim result (Task 7).

Low frequency (C_π, C_μ open). C_1 and C_2 each give a zero at $s = 0$. C_1 and C_E interact through the base input impedance, so their poles are the roots of a second-order denominator (solved exactly for Table 2; single-capacitor estimates shown):

$$\omega_{z3} = \frac{1}{R_E C_E}, \quad \omega_{p3} \approx \frac{1}{C_E[R_E \parallel (r_\pi + R_s \parallel R_B)/(\beta+1)]},$$

$$\omega_{p2} \approx \frac{1}{C_1[R_s + R_B \parallel (r_\pi + (\beta+1)R_E)]}, \quad \omega_{p1} = \frac{1}{C_2(R_C \parallel r_o + R_L)}$$

High frequency (C_1, C_2, C_E shorted). At this bias the 2N3904 model gives $C_\mu = (3.64\text{ pF})/(1+6.3/0.75)^{0.31} \approx 1.8$ pF and $C_\pi = g_m\tau_F + C_{je} \approx 12.2 + 6.4 \approx 19$ pF ($f_T \approx 310$ MHz). With $R_X = (r_b + R_s \parallel R_B) \parallel r_\pi \approx 10\ \Omega$ seen at the internal base:

$$A_v(s) = A_m \frac{1 - sC_\mu/g_m}{1 + b_1 s + b_2 s^2}$$

$$b_1 = R_X\left[C_\pi + (1 + g_m R'_L)C_\mu\right] + R'_L C_\mu = 8.16\text{ ns}, \quad b_2 = R_X R'_L C_\pi C_\mu = 1.07 \times 10^{-18}\text{s}^2$$

The roots give $f_{p4} = 19.8$ MHz (close to $1/(2\pi b_1)$) and $f_{p5} = 1.19$ GHz; the right-half-plane zero is at $g_m/(2\pi C_\mu) = 3.57$ GHz.

Table 2: Poles and zeros of $A_v(s) = V_{out}(s)/V_{in}(s)$: hand estimates and MultiSim values.

	Origin	s **(rad/s)**	f	**MultiSim** f
z_1, z_2	C_1, C_2 block DC	0	0 Hz	
z_3	C_E with R_E: $1/(R_E C_E)$	-23.3	3.70 Hz	
z_4	C_μ feedforward: g_m/C_μ (RHP)	$+2.24 \times 10^{10}$	3.57 GHz	
p_1	C_2 with $R_C \parallel r_o + R_L$	-2.13×10^{-2}	3.38 mHz	
p_2	C_1	-12.9	2.06 Hz	
p_3	C_E (dominant low-frequency pole)	-4.11×10^3	654 Hz	
p_4	Miller C_μ and output node (dominant high)	-1.25×10^8	19.8 MHz	
p_5	Non-dominant high-frequency pole	-7.47×10^9	1.19 GHz	

On the bench. The generator's $R_s = 50\ \Omega$ raises R_X to about 59 Ω, lowering f_{p4} to about 8 MHz. Scope cable and probe capacitance C_L at the collector then dominates: $f_H \approx 1/(2\pi R'_L C_L) \approx 0.5$ MHz for $C_L \approx 100$ pF, so expect a measured upper cutoff far below the MultiSim value. f_L stays near 650 Hz.

Task 8: Bode plot

Straight-line slopes: +20 dB/dec up to p_2 (2.06 Hz), flat to z_3 (3.70 Hz), +20 dB/dec to p_3 (654 Hz), flat at 42.1 dB to p_4 (19.8 MHz), then −20 dB/dec, −40 dB/dec after p_5 and −20 dB/dec after z_4.

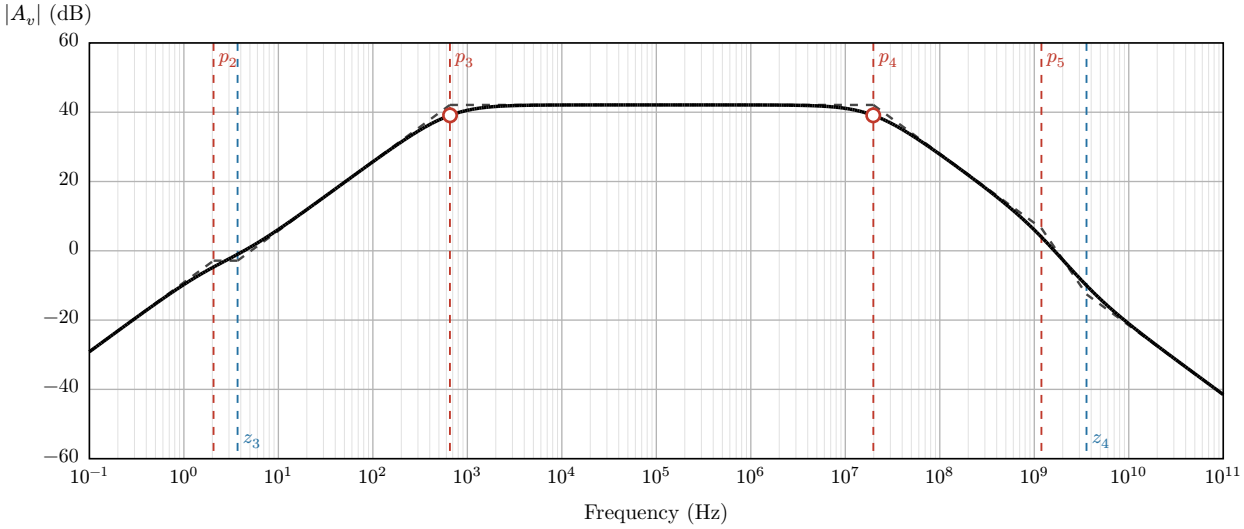


Figure 3: Voltage-gain magnitude from Table 2: exact (solid) and straight-line asymptotes (dashed). Red and blue dashed lines mark poles and zeros ($p_1 = 3.4$ mHz is off scale); circles mark the 3-dB points, $f_L \approx 655$ Hz and $f_H \approx 19.7$ MHz.